Solution

We have,

$$f(x) = \log(1+x) - \frac{x}{1+x}$$

$$\text{Therefore, } f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2}$$

$$= \frac{x}{(1+x)^2}.$$

Hence,

$$f'(x) \text{ is } \begin{cases} < 0 & \text{when } -1 < x < 0 \\ = 0 & \text{when } x = 0 \\ > 0 & \text{when } x > 0 \end{cases}$$

Therefore $f(x)$ is strictly increasing for $x > 0$ and strictly decreasing for $x < 0$. Since $f'(x)$ changes from negative to positive when passing through $x = 0$, the first derivative test tells us there is a local minimum at $x = 0$ which is $f(0) = 0$. Further, for $x > 0$, $f(x) > f(0) = 0$ gives

$$\log(1+x) - \frac{x}{1+x} > 0 \Rightarrow \log(1+x) > \frac{x}{1+x} \text{ on } (0,\infty).$$

Example 7.54

Find the intervals of monotonicity and local extrema of the function $f(x) = x\log x + 3x$.

Solution

The given function is defined and is differentiable at all $x \in (0,\infty)$.

$$f(x) = x\log x + 3x.$$

$$\text{Therefore } f'(x) = \log x + 1 + 3 = 4 + \log x.$$

The stationary numbers are given by $4 + \log x = 0$.

$$\text{That is } x = e^{-4}.$$

Hence the intervals of monotonicity are $(0, e^{-4})$ and (e^{-4}, ∞).

At $x = e^{-5} \in (0, e^{-4})$, $f'(e^{-5}) = -1 < 0$ and hence in the interval $(0, e^{-4})$ the function is strictly decreasing.

At $x = e^{-3} \in (e^{-4}, \infty)$, $f'(e^{-3}) = 1 > 0$ and hence strictly increasing in the interval (e^{-4}, ∞). Since $f'(x)$ changes from negative to positive when passing through $x = e^{-4}$, the first derivative test tells us there is a local minimum at $x = e^{-4}$ and it is $f(e^{-4}) = -e^{-4}$.

Example 7.55

Find the intervals of monotonicity and local extrema of the function $f(x) = \frac{1}{1+x^2}$.

Solution

The given function is defined and is differentiable at all $x \in (-\infty,\infty)$. As

$$f(x) = \frac{1}{1+x^2},$$

$$\text{we have } f'(x) = -\frac{2x}{(1+x^2)^2}.$$

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